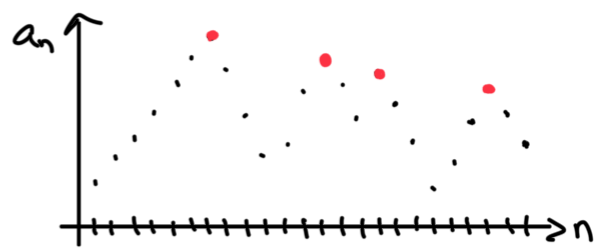
Prop For any sequence, there is a subsequence that is monotone.

Pf: Let $\{a_n\}_{n\geq 1}$ be a sequence.

We will call the nth term "dominant" if

$$a_m < a_n \quad \forall m > n.$$

Case 1: There are infinitely many dominant terms. Then the subsequence formed by these dominant terms is strictly decreasing.



Case 2: There are finitely many dominant terms. Let N be larger than the largest index of the dominant terms. So $\forall n \geq N$, a_n is not dominant.

- Set $k_1 = N$, so that $a_{k_1} = a_N$
- As a_{k_1} is not dominant, $\exists k_2 > k_1$ s.t. $a_{k_2} \geq a_{k_1}$.
- As $k_2 > k_1 = N$, then a_{k_2} is not dominant, so $\exists\, k_3 > k_2$ s.t. $a_{k_3} \geq a_{k_2}$.

Proceeding inductively, we construct a subsequence $\{a_{k_n}\}_{n\geq 1}$ s.t.

$$a_{k_{n+1}} \geq a_{k_n} \quad \forall n \geq 1. \qquad \square$$

Thm (Bolzano-Weierstrass) For any bounded sequence, there exists a subsequence that converges.

<u>Pf</u>: Let $\{a_n\}_{n\geq 1}$ be a bounded sequence. By the previous proposition, $\exists$ a subsequence $\{a_{k_n}\}_{n\geq 1}$ that is monotone. As $\{a_n\}_{n\geq 1}$ is bounded, so is $\{a_{k_n}\}_{n\geq 1}$. Together, we conclude $\{a_{k_n}\}_{n\geq 1}$ is monotone and bounded, and therefore converges (by Lecture 8). $\square$

- To finish the proof of the Cauchy criterion:

<u>Prop</u> If $\{a_n\}_{n\geq 1}$ is a Cauchy sequence, then it converges.

<u>Pf</u>: Step 1: $\{a_n\}_{n\geq 1}$ is bounded (by Lecture 8).

Step 2: $\exists$ a subsequence $\{a_{k_n}\}_{n\geq 1}$ which converges (by Bolzano–Weierstrass).

Step 3: The whole sequence $\{a_n\}_{n\geq 1}$ converges.

Set $a = \lim_{n\to\infty} a_{k_n}$. Fix $\varepsilon > 0$. Then $\exists N_1 \in \mathbb{N}$ s.t.

$$|a_{k_n} - a| < \frac{\varepsilon}{2} \quad \forall n \geq N_1 .$$

As $\{a_n\}_{n\geq 1}$ is Cauchy, $\exists N_2 \in \mathbb{N}$ s.t.

$$|a_n - a_m| < \frac{\varepsilon}{2} \quad \forall\, n, m \geq N_2 .$$

Set $N = \max\{N_1, N_2\}$. Then, for $n \geq N$ we have

$$|a_n - a| \leq |a_n - \underbrace{a_{k_n}}_{k_n \geq n \geq N \geq N_2}| + |a_{k_n} - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon .$$

So $\lim_{n\to\infty} a_n = a$. $\square$

- This finishes the proof of the Cauchy criterion.

Def Let $\{a_n\}_{n\geq 1}$ be a sequence. We say:

① $\{a_n\}_{n\geq 1}$ <u>diverges to $+\infty$</u> if $\forall M>0$ $\exists N\in\mathbb{N}$ s.t. $a_n > M$ $\forall n\geq N$. When this is true, we write $\lim_{n\to\infty} a_n = +\infty$.

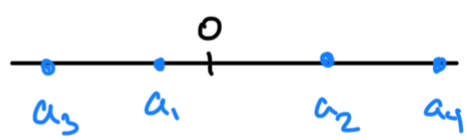
② $\{a_n\}_{n\geq 1}$ <u>diverges to $-\infty$</u> if $\forall m<0$ $\exists N\in\mathbb{N}$ s.t. $a_n < m$ $\forall n\geq N$. When this is true, we write $\lim_{n\to\infty} a_n = -\infty$.

- Recall: "diverge" = "does not converge"

Ex ① $\{\sqrt{n}\}_{n\geq 1}$ diverges to $+\infty$.

② $\{(-1)^n\}_{n\geq 1}$ diverges, but is bounded.

③ $\{(-1)^n n\}_{n\geq 1}$ diverges and is unbounded, but does not diverge to $+\infty$ or $-\infty$.



Prop If $\{a_n\}_{n\geq 1}$ is unbounded and $\begin{cases}\text{increasing}\\ \text{decreasing}\end{cases}$ then $\{a_n\}_{n\geq 1}$ diverges to $\begin{cases}+\infty\\ -\infty\end{cases}$ respectively.

Rmk If a sequence is increasing, there are only 2 possibilities:

- bounded $\Rightarrow$ converges
- unbounded $\Rightarrow$ diverges to $+\infty$

<u>Pf</u>: Case: $\{a_n\}_{n\geq 1}$ is increasing. Let $M>0$. As $\{a_n\}_{n\geq 1}$ is bounded below by a_1, it must be unbounded above. Then M cannot be an upper bound, and so $\exists N \in \mathbb{N}$ s.t. $a_N > M$. So, for $n \geq N$,

$$a_n \geq a_N > M.$$

Therefore $\lim_{n\to\infty} a_n = +\infty$.

Case: $\{a_n\}_{n\geq 1}$ is decreasing. Apply previous case to $\{-a_n\}_{n\geq 1}$. (Exercise.) $\square$

<u>Rmk</u> "$\lim_{n\to\infty} a_n$" only sometimes makes sense:

① converges $\Rightarrow$ $\lim_{n\to\infty} a_n = a \in \mathbb{R}$

② diverges $\begin{cases} \text{diverges to } \pm\infty \Rightarrow \lim_{n\to\infty} a_n = \pm\infty \\ \text{else} \Rightarrow \lim_{n\to\infty} a_n \text{ does not exist} \end{cases}$

- It would be nice to have a tool that <u>always</u> makes sense...

LIMSUP AND LIMINF

Let $\{a_n\}_{n\geq 1}$ be a sequence of real numbers (convergent or not).

<u>Case 1</u>: $\{a_n\}_{n\geq 1}$ is bounded. Fix any $N \in \mathbb{N}$. Then the set $\{a_n : n \geq N\}$ is nonempty and bounded, so

$$u_N = \inf\{a_n : n \geq N\}, \qquad v_N = \sup\{a_n : n \geq N\}$$

both exist. We automatically have $u_N \leq v_N \ \forall N$.

Claim: $\{u_N\}_{N \geq 1}$ is increasing and $\{v_N\}_{N \geq 1}$ is decreasing. (Next time!)